

UNIT 2

Ecosystems

"Everytime we lose a species we break a life chain which has evolved over 3.5 billion years".

-Jeffrey McNeely, IUCN Chief Scientist

Objectives

- Introduction
- Structure and function of an ecosystem
- Energy Flow through the Ecosystem
- Ecological succession
- Food chains, food webs and ecological pyramids
- Types, Characteristic Features, Structures and Function of the Ecosystem

2.1 INTRODUCTION

An **ecosystem** is a community of living organisms (plants, animals and microbes) that occurs in some locale, in conjunction with physical and chemical factors (air, water and soil, climate etc.). These features create conducive conditions to make the plants and animals live. The biotic and abiotic components are linked together through nutrient cycles and energy flows. Ecosystems are generally fragile and can be disturbed by human activities like mountains are disturbed by constructions of roads; buildings and rivers are disturbed by dams and so on. Ecosystems can be classified as natural ecosystem which includes the forests, grasslands, deserts, ponds, rivers, lakes, and the sea and man-modified ecosystems which include agricultural land and urban or industrial land use patterns.

2.1.1 Components of Ecosystems

The classification of ecosystem into biotic and abiotic component is given in Table 2.1.

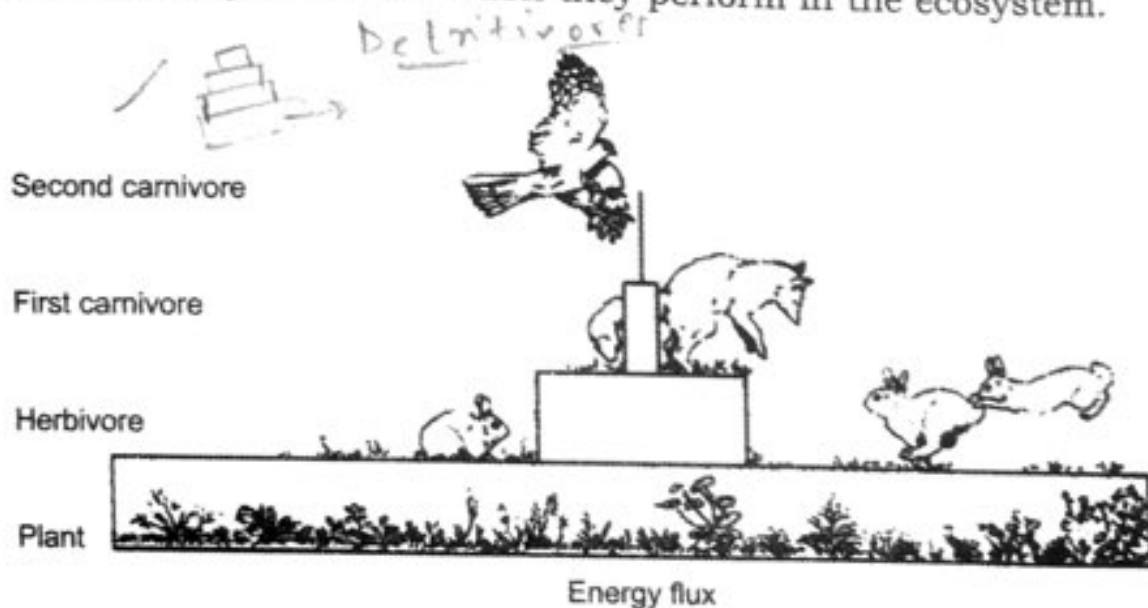
Table 2.1 Biotic and Abiotic Components

Abiotic Components	Biotic Components
Sunlight Temperature Precipitation Water or moisture Soil or water chemistry (e.g., P, NH_4^+) etc.	Primary producers Herbivores Carnivores Omnivores Detritivores etc.

2.2 STRUCTURE AND FUNCTION OF AN ECOSYSTEM

The two main structural features of any ecosystem are its 'species composition' and 'stratification'. This is also called the 'Trophic structure' of an ecosystem, where each animal population is present in different trophic level. Plants or the producers are in the first trophic level, Herbivores are in the second trophic level because they feed on producers followed by secondary consumers, tertiary consumers and so on.

Ecosystem processes are broad generalizations that actually take place through the actions of individual organisms. The nature of the organisms, functional groups and trophic levels to which they belong decide the efficiency of the activities of these individuals. A **functional group** is a biological category composed of organisms that perform mostly the same kind of function in the system. Plants are the producers in the ecosystem as they manufacture their food by using energy from the sun. Hence all the photosynthetic plants are in the same functional group. Members of the same functional group do not depend on the identity of the individual members but only on the function they perform in the ecosystem.

**Fig. 2.1** Structure of an ecosystem

2.2.1 Producers, Consumers and Decomposers

The biotic community can be divided into producers, consumers and decomposers. Producers are the green plants or algae which can synthesize glucose (organic compound) from carbon dioxide and water (inorganic

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Sunlight	Primary producers
Temperature	Herbivores
Precipitation	Carnivores
Water or moisture	Omnivores
Soil or water chemistry (e.g., P, NH_4^+)	Detritivores
etc.	etc.

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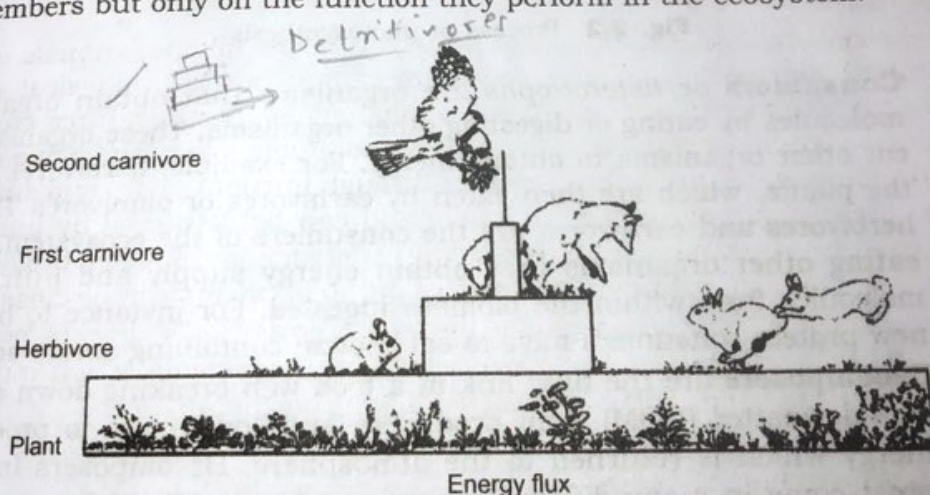


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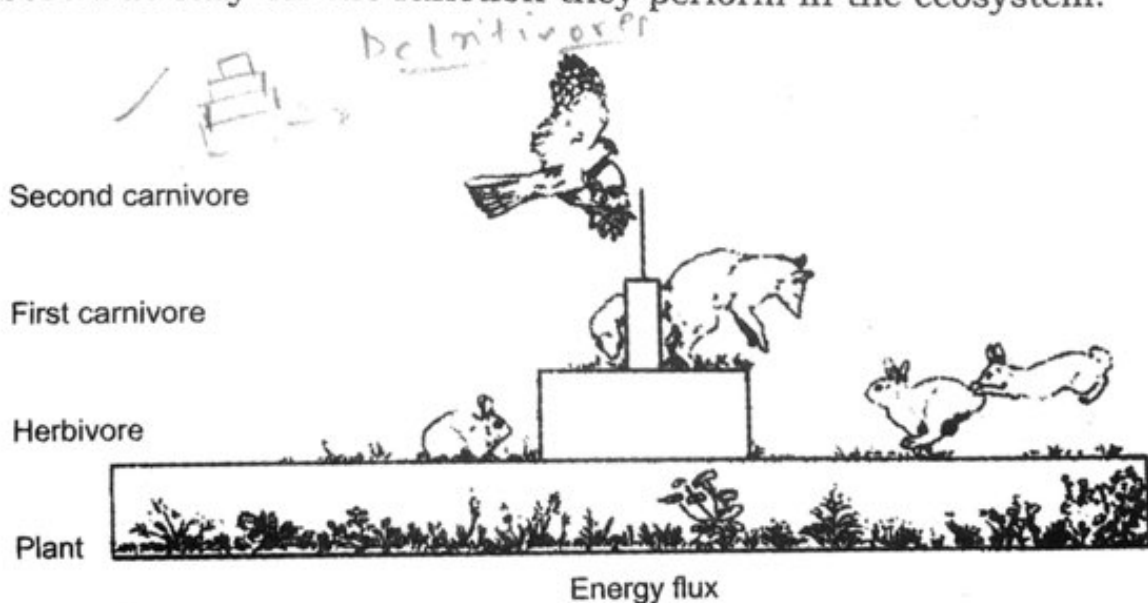
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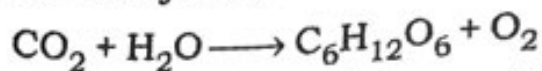
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2.2.1 Producers, Consumers and Decomposers

The biotic community can be divided into producers, consumers and decomposers. Producers are the green plants or algae which can synthesize glucose (organic compound) from carbon dioxide and water (inorganic

substance) in the presence of sunlight. The glucose so produced can be used both as a direct energy source or can combine with other molecules to produce biomass that provides the total theoretical energy available to all non photosynthesising organisms in the ecosystem. The primary function of an ecosystem is a systematic transfer of energy from one organism to another. This energy flow drives the movement of vital nutrients throughout the ecosystem



Producers transform light energy (from the sun) into chemical energy

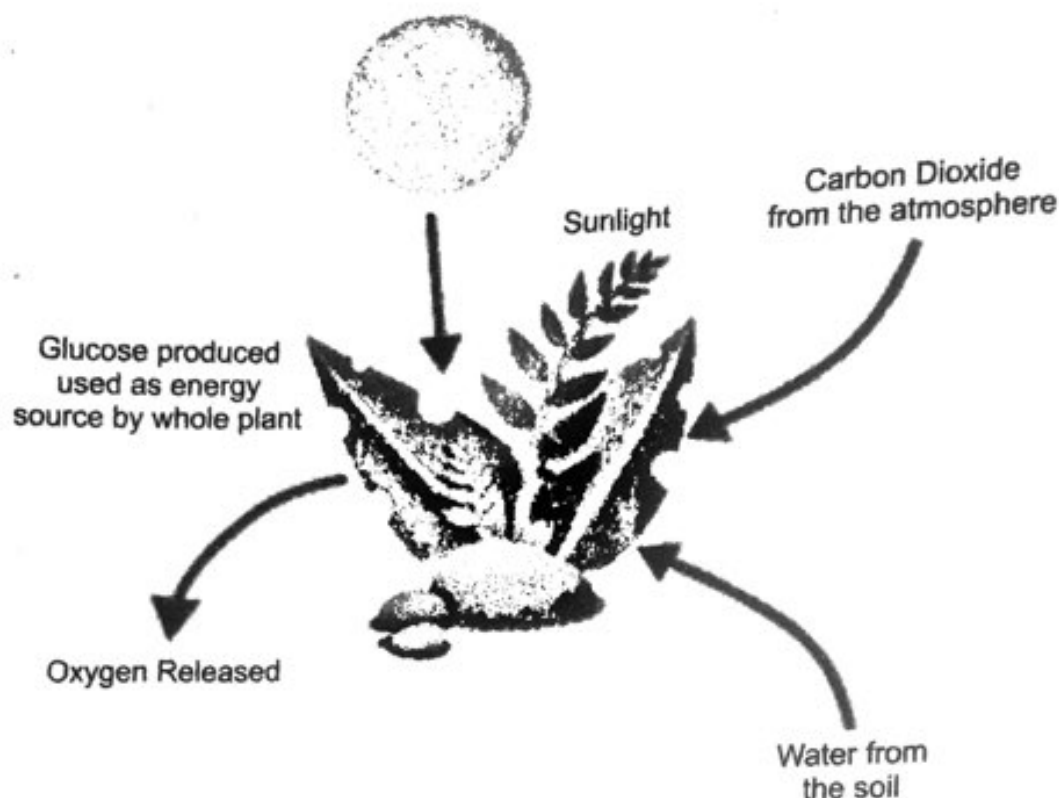


Fig. 2.2 Process of photosynthesis

- **Consumers** or *heterotrophs* are organisms that obtain organic molecules by eating or digesting other organisms. These organisms eat other organisms to obtain energy. For example, herbivores eat the plants, which are then eaten by carnivores or omnivores. Thus herbivores and carnivores are the consumers of the ecosystem. By eating other organisms they obtain energy supply and nutrient molecules from within the biomass ingested. For instance to build new protein consumers have to eat protein containing amino acids.
- **Decomposers** are the final link in a food web breaking down dead organic matter (DOM) from producers and consumers to produce energy which is returned to the atmosphere. Decomposers in the forest come in many different shapes and sizes. Shelf fungus is a typical fungus that grows on the sides of trees. It grows into the tree and decomposes it slowly.

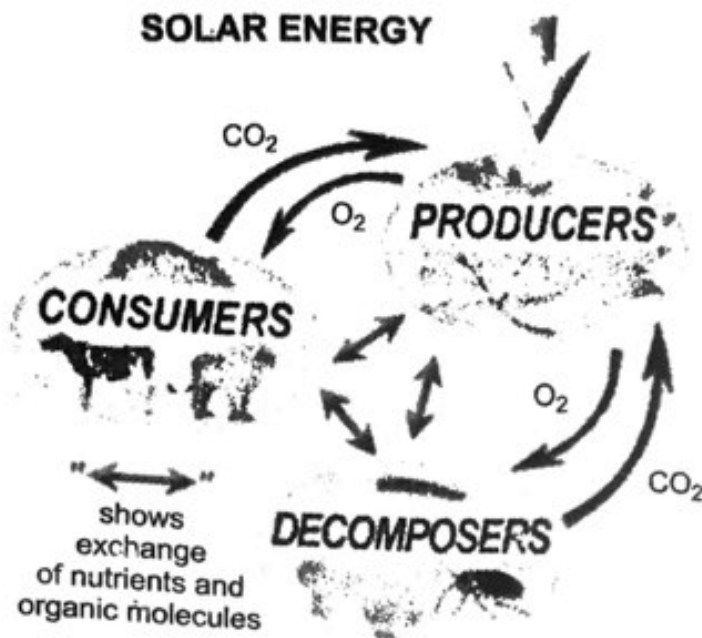


Fig. 2.3 Producers, Consumers and Decomposers in a ecosystem

2.3 ENERGY FLOW THROUGH THE ECOSYSTEM

In the ecosystem sun is the largest source of energy. In the food chain energy flows from one organism to the other when one organism eats another organism. Any energy remaining in a dead organism is consumed by decomposers. Nutrients can be cycled through an ecosystem but energy is simply lost over time. In ecology, **energy flow**, also called the **calorific flow**, refers to the flow of energy through a food chain. In an ecosystem, ecologists seek to quantify the relative importance of different component species and feeding relationships.

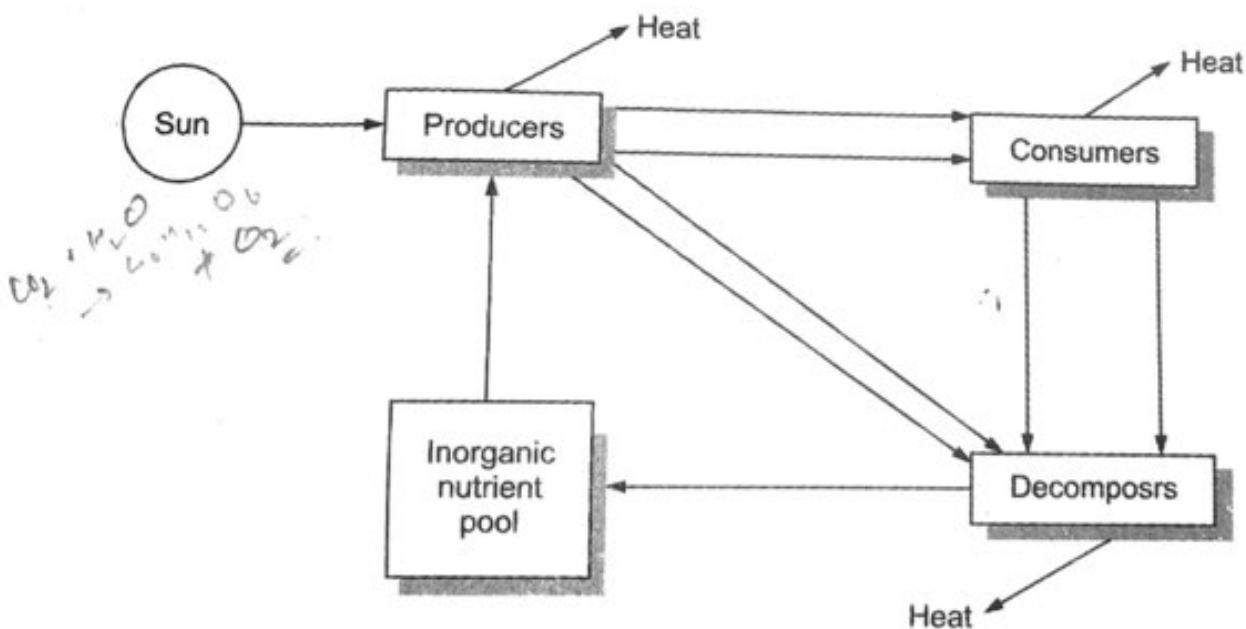
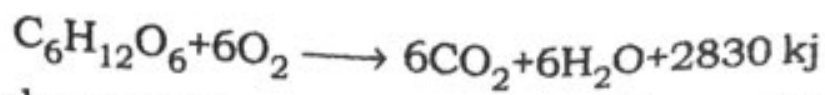


Fig. 2.4 Energy Flow Diagram

The diagram above shows the flow of both energy and inorganic nutrients through the ecosystem. When respiration occurs, the carbon-carbon bonds are broken and the carbon oxygen bonds are formed to produce carbon dioxide. During plant respiration, a carbohydrate combines with oxygen and is reduced to carbon dioxide, water, and heat.



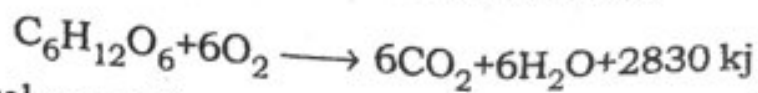
This process releases the energy, which is either used by the organism (to move its muscles, digest food, excrete wastes, think, etc.) or the energy may be lost as heat. The dark arrows represent the movement of this energy Solar Energy that is not used in an ecosystem is eventually lost as heat. The other component shown in the diagram is the inorganic nutrients (which do not contain carbon-carbon bonds. The movement of the inorganic nutrients is represented by the open arrows. The autotrophs obtain these inorganic nutrients from the inorganic nutrient pool, which is usually the soil or water surrounding the plants or algae. . In the food chain energy flows from one organism to the other, when one organism eats another organism. Any energy remaining in a dead organism is consumed by decomposers. Nutrients can be cycled through an ecosystem but energy is simply lost over time. The inorganic nutrients are recycled, the energy is not.

2.4 ECOLOGICAL SUCCESSION

Ecological succession is the observed process of change in the species structure of an ecological community over time. The community begins with relatively few living organisms (plants and animals) and becomes complex over a period of time and becomes stable. Sometimes, within a community some species may become less and eventually vanish due to some reason. Ecological succession occurs because within a ecosystem an organism can have optimum growth and produce the most viable offspring till set of environmental conditions remains constant. If there is any change in the environment, the previously dominant species may fail and another species may become ascendant thus giving rise to economic succession.

Ecological Succession		Year	
	Crab-grass	0-1	
	Tall grass-herbaceous plants	1-3	
	Pines come in	3-10	
	Pine forest	10-30	
	Hard-woods come in	30-70	
	Hard-wood forest climax	70+	

Fig. 2.5 Ecological succession



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$2 + 6H_2O + 2830 \text{ kJ}$
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The most often quoted examples of succession deal with plant succession. It is worth remembering that as plant communities' change, so will the associated microorganism, fungus and animal species. Succession involves the **whole** community, not just the plants. Change in the plant species present in an area is one of the driving forces behind changes in animal species. This is because each plant species will have associated animal species which feed on it. The presence of these herbivore species will then dictate which particular carnivores are present.

2.5 FOOD CHAINS, FOOD WEBS AND ECOLOGICAL PYRAMIDS

- **The food chains:** In nature energy must flow from one living organism to another. Energy is transferred from plants to animals when herbivores feed on plants and subsequently from one animal to the other in case of carnivores. In an ecosystem, some of the animals feed on other living organisms, while some feed on dead organic matter. The latter form the 'detritus' food chain. At each linkage in the chain, a major part of the energy from the food is lost for daily activities. Each chain usually has only four to five such links. However a single species can be linked with a large number of species.
- **The food webs:** In the ecosystem, a large number of interlinked chains constitute a web. If due to any human activity any link in the web is broken, then the species linked with the web may become extinct.
- An **ecological pyramid** (also **trophic pyramid** or **energy pyramid**) is a graphical representation designed to show the biomass (organic matter present in an organism) or biomass productivity (production of biomass) at each trophic level in a given ecosystem.

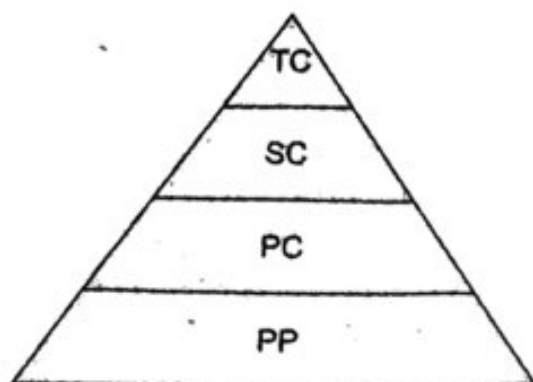
Ecological pyramids begin with producers at the bottom (such as plants) and proceed through the various trophic levels (such as herbivores that eat plants, then carnivores that eat herbivores, then carnivores that eat other carnivores, and so on). The highest level is the top of the food chain. This pyramid actually refers to the amount of biomass present in unit area (expressed in grams per meter² or calories per meter²). One gets a similar shape, whether the food or energy relationship between organism at different trophic level is expressed. Thus, relationship is expressed in terms of number, biomass or energy. The base of each pyramid represents the producers or the first trophic level while the apex represents tertiary or top level consumer. The three ecological pyramids that are usually studied are (a) pyramid of number; (b) pyramid of biomass and (c) pyramid of energy. To show the total biomass of individuals in each trophic level. Pyramid of biomass is better than pyramid of number in showing the relationships between organisms. Pyramid of biomass is upright in case of grassland and forest ecosystem whereas inverted in case of pond

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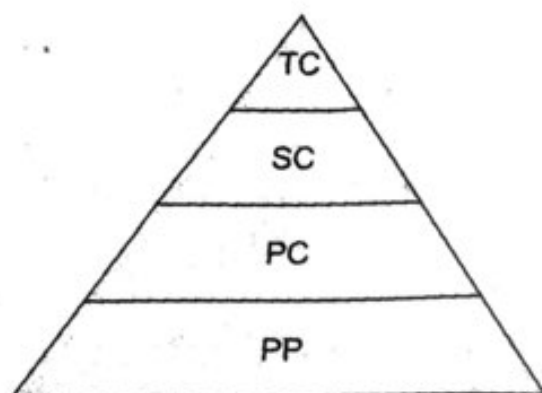
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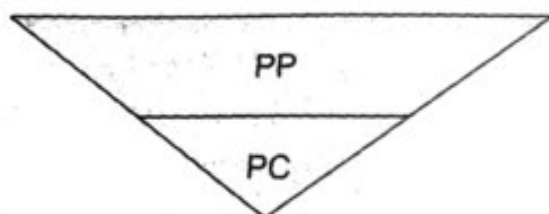
Trophic Level	No. of Individuals
TC (Tertiary Consumer)	3
SC (Secondary Consumer)	3,54,000
PC (Primary Consumer)	7,08,000
PP (Primary Producer)	5,842,000

Fig. 2.6 Pyramid of numbers in a grassland ecosystem. Only three top carnivores are supported in an ecosystem based on production of nearly 6 millions plants (Source: ncert.nic.in)



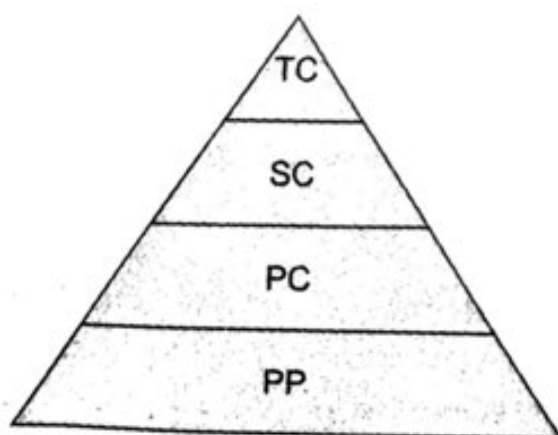
Trophic Level	Dry Weight (Kg/m ³)
TC (Tertiary Consumer)	1.5
SC (Secondary Consumer)	11
PC (Primary Consumer)	37
PP (Primary Producer)	809

Fig. 2.7 Pyramid of biomass shows a sharp decrease in biomass at higher trophic levels. (Source: ncert.nic.in)



Trophic Level	Dry Weight (Kg/m ³)
PP (Primary Producer)	21
PC (Primary Consumer)	4

Fig. 2.8 Inverted pyramid of biomass small standing crop of phytoplankton supports large standing crop of zooplankton (Source: ncert.nic.in)



1,000,000 J Sunlight

Trophic Level	Energy (J)
TC (Tertiary Consumer)	10
SC (Secondary Consumer)	100
PC (Primary Consumer)	1000
PP (Primary Producer)	10,000

Fig. 2.9 An ideal pyramid of energy. Observe that primary producers convert only 1% of the energy in the sunlight available to them into NPP (Source: ncert.nic.in)

2.6 TYPES, CHARACTERISTIC FEATURES, STRUCTURES AND FUNCTION OF THE ECOSYSTEM

The ecosystem can be divided into two major groups:

2.6.1 Terrestrial Ecosystems

This category of ecosystem includes, Grassland, Semi arid areas, Deserts, Mountains and Islands.

2.6.2 Aquatic Ecosystems

This category of ecosystem includes ponds, streams, lakes, rivers, oceans, estuaries

Terrestrial ecosystems in this type of ecosystem since land is an integral part, it has been modified into agricultural, pastoral or forest area. In the recent times the land has been rapidly converted into irrigated agricultural ecosystems or urban and industrial centers. The conversion has led to the overuse or misuse of land thereby degrading the environment.

(i) Forest ecosystem: A forest is an ecosystem that supports innumerable plant and animal species as well as ground cover, water, and air subsystems. The subsystems which are also the by products of the forest, provide the essence of life. Each forest type forms a habitat for a specific community of animals that are adapted to live in it. The forest has a vertical structure, horizontal structure and environmental structure.

Vertical structure: The forest is divided into distinct layers, each adapted to increasingly filtered sunlight if going top down. The layers are: canopy, understory, ground layer, and the forest floor. Not all forests have each layer. Generally; extreme environments such as deeply shaded woodlands, boreal forests, and frequently flooded woodlands have fewer layers.

- **Horizontal Structure:** The composition and distribution of species can vary widely from one place to the other depending on the environmental gradients, such as moisture, drainage, slope, aspect, soil type, and light intensity
- **Environmental structure:** The forest is influenced by environmental factors like soil type, temperature, moisture, precipitation pattern etc.

The classification of forest is given in the following Table 2.2.

(ii) Grassland ecosystem

Grasslands are areas where the vegetation is dominated by grasses. The vegetation is shrubs, plants, woody trees etc. Grasslands are predominantly found in areas where the temperature varies between -5°C to 20°C and the rainfall is between 600-1500 mm. The vegetation present in the grassland can be both natural and cultivated. The most predominant grassland of India is found in the floodplains of rivers coming out of the Himalayas onto the Indo-Gangetic Plain — mainly composed of alluvial

Table 2.2 Classification of Forests

Forest type	Plants Examples	Common Animal Examples
Himalayan Coniferous	Pine, Deodar	Wild goats and sheep, Himalayan black bear
Himalayan Broadleaved	Maple, Oak	Himalayan Tahr, Asiatic Black Bear, common Leopard
Evergreen North-east, western Ghats, Andaman & Nicobar	Jamun, Ficus, Dipterocarpus	Tiger, Leopard, Sambar, Malabar whisting thrush, Malabar Pied hornbill, tree frogs.
Deciduous- Dry	Teak, Ain, Terminalia	Tiger, Chital, Barking deer, Babblers, Flycatchers, Hornbills
Thom and scrub, Semiarid forests	Babul, Ber, Neem	Blackbuck, Chinkara, Fourhorned antelope, Partridge, Monitor lizard
Mangrove Delta Forest	Avicenia	Crocodile, shorebirds- sandpipers, plovers, fish, crustacean.

*Source: www.ugc.ac.in

sediments — between the Yamuna and Brahmaputra drainages. This is known as the **Terai Duar Grassland**. A major part of the grassland is destroyed by man for agriculture. The Terai-Duar floodplain grasslands has huge populations of one-horned rhinoceros, Asian elephant, gaur, nilgai, water buffalo, an impressive diversity of deer, pygmy hogs and Bengal tigers.

(iii) Lowland Grassland

There are grasslands which are also found on the fringes of lowland desert and semidesert. An example of such kind is the Rann of Kutch in northwestern India, where the Luni River descends out of the Aravalli Hills to a complex of desert and seasonal salt marsh inland of the Gulf of Kachchh. The predominant animals found here are large mammals, most notably the Asiatic wild ass, blackbuck, nilgai and chinkara, Indian wolf, striped hyena and caracal. The park's open country also provides critical habitat for other grassland specialists, such as the lesser florican and harrier hawks, which gather here in their greatest global roost.

(iv) Desert ecosystem

A **desert** is a barren area of land with scanty rainfall and hostile living conditions for plant and animal life. About one third of the land surface of the world is arid or semiarid. This includes much of the polar regions where little precipitation occurs and which are sometimes called "cold deserts". In India, cold deserts are found in Ladakh (high plateau of Himalayas). There are a number of ways of classifying deserts including by the amount of precipitation that falls, by the temperature that prevails, by the causes of desertification or by the geographical location of the desert.

Although rainfall in the deserts is scarce, there are occasional downpours that can result in flash floods. There are frequent dust storms which result in the formation of sand dunes. In India such type of desert is seen in that desert of Rajasthan. There are also areas covered with sparse grasses and a few shrubs, which grow if it rains.

The Great and Little Rann of Kutch are highly specialised arid ecosystems. In the summers the landscape is similar to that of a desert but as this place is near to the sea, it gets converted into salt marshes during the monsoons and attract aquatic birds such as ducks, geese, cranes, storks, etc. The Great Rann is famous, as it is the only known breeding colony of the Greater and Lesser Flamingos in our country. The Little Rann of Kutch is the only home of the wild ass in India. Desert and semi arid regions have a number of highly specialized animals. Animals need to keep cool and find enough food and water to survive. Many are nocturnal and stay in the shade or underground during the heat of the day. They tend to be efficient at conserving water, extracting most of their needs from their food and concentrating their urine. The rare animals include the Indian wolf, desert cat, desert fox and birds such as the Great Indian Bustard and the Florican.

An **aquatic ecosystem** is an ecosystem of water body. Plants and animals that can survive in the environment of water live in aquatic ecosystems. Aquatic ecosystems are divided into two main categories that is marine ecosystems and freshwater ecosystems

(i) Marine Ecosystem

Marine ecosystems cover approximately 71% of the Earth's surface and contain approximately 97% of the planet's water. They are distinguished from freshwater ecosystems by the presence of dissolved compounds, especially salts (mainly sodium chloride) giving an average salinity of 35 parts per thousand (ppt) of water. Actual salinity varies among different marine ecosystem. Ecosystems.

Marine ecosystems can be divided into many zones depending upon water depth and shoreline features. The oceanic zone is the vast open part of the ocean where animals such as whales, sharks, and tuna live and near-shore zones can include estuaries, salt marshes, coral reefs, lagoons and mangrove swamps the base of the food web.

(ii) Estuaries

An estuary is a unique environment where fresh water and saltwater come together. Estuaries are generally found on the coast where a river or bay or wherever the river can enter the sea. Since estuaries are affected by the tides hence salinity, temperature and other physical properties in an estuarial system can change making the organisms tolerant to change. Even with that consideration, estuaries are amongst the most fertile places in the world, where all kinds of plants and animals live.

Freshwater ecosystems cover 0.80% of the Earth's surface and inhabit 0.009% of its total water. Freshwater ecosystems contain 41% of the world's known fish species and can be subdivided into:

- Lentic: slow moving water, including pools, ponds, and lakes.
- Lotic: faster moving water, for example streams and rivers.
- Wetlands: areas where the soil is saturated or inundated for at least part of the time.

(iii) Lake Ecosystem

A lake is a body of water completely surrounded by land and can have either saline or fresh water. Lakes are not permanent because the water can both sink into the ground and evaporate. In order for a lake to remain, it must be constantly fed by a river or rainfall. Plants which are mainly found in lakes is algae, which derives energy from the sun. There are fish that are herbivorous and are dependent on algae and aquatic weeds. The small animals such as snails are used as food by small carnivorous fish, which in turn are eaten by larger carnivorous fish.

(iv) Streams and Rivers

Streams and rivers are flowing water ecosystems in which all the living forms are specially adapted to different rates of flow. Rivers which generally come from mountain springs, melting glaciers or lakes, can either flow on land or underground like in deserts and seas. A river's contribution to the water cycle is that it collects water from the ground and returns it to the ocean. The major zones in river ecosystems are determined by the river bed's gradient or by the velocity of the current. Faster moving turbulent water typically contains greater concentrations of dissolved oxygen, and hence supports more biodiversity. Upland and lowland rivers are the result of distinction of the velocity gradient. Some plants and animals such as snails and other burrowing animals can withstand the rapid flow of the hill streams. Other species of plants and animals such as water beetles and skaters can live only in slower moving water. Some species of fish, such as Mahseer, go upstream from rivers to hill streams for breeding. They need crystal clear water to be able to breed. They lay eggs only in clear water so that their young can grow successfully. As deforestation occurs in the hills the water in the streams that once flowed throughout the year become seasonal. This leads to flash floods in the rains and a shortage of water once the streams dry up after the monsoon. Environmental threats to rivers include loss of water, dams, chemical pollution and introduced species.

(v) Ponds

Ponds are small bodies of freshwater water over 1 m square that is inundated, "wet" for more than 8 months of the year. They have shallow and still water, can be marshy, and support aquatic plants. The size and depth of ponds often varies greatly with the time of year. Many ponds are produced by spring flooding from rivers. A large number of plants and animals live in ponds giving rise to a food web and biodiversity. There is usually a diverse array of aquatic life, with a few examples including algae, snails, fish, beetles, water bugs, frogs, turtles and sometimes even alligators.

Ecosystems

EXERCISES

1. Explain the concept of an ecosystem with their structure and function.
2. What do you understand by Ecological succession?
3. Explain with the help of the diagram the energy flow in an ecosystem?
4. Why do tertiary consumers occur at a low density in an ecosystem?
5. What are the two components of an ecosystem. Give examples.
6. With the help of examples explain producers, consumers and decomposers.
7. What do you understand by food chain and food web? How does the interlinks between the food web break.
8. Explain ecological pyramid. Is it different from energy pyramid.
9. How does the nutrients and the energy flow from one organism to the other
10. Why our grassland species are vanishing? What will be its consequences?
11. What is grassland ecosystem? What are the threats to grassland ecosystem?
12. What are the threats to forest ecosystem? How forest ecosystem can be conserved?
13. What is a desert ecosystem? How can desert ecosystem be conserved?
14. How does an aquatic ecosystem support a wide range of biodiversity?



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UNIT 3

Biodiversity and its Conservation

"It is reckless to suppose that biodiversity can be diminished indefinitely without threatening humanity itself".

– Prof. Edward O. Wilson "Father of Biodiversity"

Objectives

- Introduction
- Biogeographical Classification of India
- Value of Biodiversity
- Biodiversity at Global, National and Local Levels
- India as a mega-diversity Nation
- Hot-spots of biodiversity
- Threats of Biodiversity
- Endangered and Threatened Species of India
- Endemic Species of India
- Biodiversity Conservation

3.1 INTRODUCTION

Biological diversity means that variety of plants and animals and other living things in a particular area or region. It also means the number or abundance of different species living within a particular region. Everything that lives in an ecosystem is part of the web of life, including humans. Each species of vegetation and each creature has a place on the earth and plays a vital role in the circle of life. Plant, animal, and insect species interact and depend upon one another for what each offers, such as food, shelter, oxygen, and soil enrichment.

Maintaining a wide diversity of species in each ecosystem is necessary to preserve the web of life that sustains all living things.

3.1.1 Three levels of biodiversity

Biodiversity refers to the variety of life on Earth at all its levels, from genes to ecosystems, and the ecological and evolutionary processes that sustain it. Biodiversity includes not only species we consider rare, threatened, or

endangered, but every living thing, even organisms, such as microbes, fungi, and invertebrates.

Biodiversity is generally understood at three levels genetic, species, and ecosystem. These levels are all interrelated yet distinct enough that they can be studied as three separate components.

Genetic Diversity

Genetic diversity is the variety present at the level of genes. Genes, made of DNA, are the building blocks that determine how an organism will develop and what its traits and abilities will be. This level of diversity can differ by alleles (different variants of the same gene, such as blue or brown eyes), by entire genes (which determine traits, such as the ability to metabolize a particular substance), or by units larger than genes such as chromosomal structure.

Genetic diversity can be measured at many different levels, including population, species, community, and biome. Which level is used depends upon what is being examined and why, but genetic diversity is important at each of these levels.

The amount of diversity at the genetic level is important because it represents the raw material for evolution and adaptation. More genetic diversity in a species or population means a greater ability for some of the individuals in it to adapt to changes in the environment. Less diversity leads to uniformity, which is a problem in the long term, as it is unlikely that any individual in the population would be able to adapt to changing conditions. As an example, modern agricultural practices use monocultures, which are large cultures of genetically identical plants. This is an advantage when it comes to growing and harvesting crops, but can be a problem when a disease or parasite attacks the field, as every plant in the field will be susceptible. Monocultures are also unable to deal well with changing conditions.

Within species, genetic diversity often increases with environmental variability, which can be expected. If the environment often changes, different genes will have an advantage at different times or places. In this situation genetic diversity remains high because many genes are in the population at any given time. If the environment didn't change, then the small number of genes that had an advantage in that unchanging environment would spread at the cost of the others, causing a drop in genetic diversity.

In communities, it can increase with the diversity of species. How much it increases depends not only on the number of species, but also on how closely related the species are. Species that are closely related (e.g. two species of neem) have similar genetic structures and makeup and therefore do not contribute much additional genetic diversity. These closely-related species will contribute to genetic diversity in the community less than more remotely-related species (e.g. neem and peepal) would.

An increase in species diversity can also affect the genetic diversity, and do so differently at different levels. If there are many species, the genetic diversity at that level will be larger than when there are fewer

Algae, snails

species. On the other hand, genetic diversity within each species can decrease. This can happen if the large number of species means so much competition that each species must be extremely specialized, such as only eating a single type of food. If they are so specialized, this specialization will lead to little genetic diversity within any of the species.

Species Diversity

Biodiversity studies typically focus on species. They do so not because species diversity is more important than the other two types, but because species diversity is easier to work with. Species are relatively easy to identify by eye in the field, whereas genetic diversity requires laboratories, time and resources to identify and ecosystem diversity needs many complex measurements to be taken over a long period of time. Species are also easier to conceptualize and have been the basis of much of the evolutionary and ecological research that biodiversity draws on.

Species are well known and are distinct units of diversity. Each species can be considered to have a particular role in the ecosystem, so the addition or loss of single species may have consequences for the system as a whole. Conservation efforts often begin with the recognition that a species is endangered in some way, and a change in the number of species in an ecosystem is a readily obtainable and easily comprehensible measure of how healthy the ecosystem is.

Ecosystem Diversity

Ecosystem-level theory deals with species distributions and community patterns, the role and function of key species, and combines species functions and interactions. The term "ecosystem" here represents all levels greater than species: associations, communities, ecosystems, and the like. One of the difficulties in examining communities is that the transitions between them are usually not very sharp. A lake may have a very sharp boundary between it and the deciduous forest it is in, but the deciduous forest will shift much more gradually to grasslands or to a coniferous forest. This lack of sharp boundaries is known as "open communities" (as opposed to "closed communities," which would have sudden transitions) and makes studying ecosystems difficult, since even defining and delimiting them can be problematic.

Some researchers think of communities as simply the sum of their species and processes, and don't think that any of the properties found in communities are special to that level. Many others disagree, claiming that many of the characteristics of communities are unique and cannot be extrapolated from the species level. Examples of these characteristics include the levels of the food chain and the species at each of those levels, guilds (species in a community that are functionally similar), and other interactions.

3.2 BIOGEOGRAPHICAL CLASSIFICATION OF INDIA

India is the seventh largest country in the world and Asia's second largest nation with an area of 3, 287, 263 sq.km. encompassing a varied landscape rich in natural resources. India is shielded by the world's highest mountains, the Himalayas, in the north. The southern part of India takes the shape of a peninsula and divides the Indian Ocean into the Bay of Bengal to the southeast and the Arabian Sea to the southwest. The southern tip of Kanyakumari is washed by the Indian Ocean. The Andaman and Nicobar Islands in the Bay of Bengal and the Lakshadweep group of islands in the Arabian sea are also a part of India.

India has a great diversity of natural ecosystems from the cold and high Himalayan ranges to the sea coasts, from the wet northeastern green forests to the dry northwestern arid deserts, different types of forests, wetlands, islands and the oceans. India consists of fertile river plains and high plateaus and several major rivers, including the Ganges, Brahmaputra and Indus. The climate of India is determined by the southwest monsoon between June and October, the northeast monsoon between October and November and dry winds from the north between December and February. From March to May the climate is dry and hot.

India, being a vast country, shows a great diversity in climate, topography and geology and hence the country is very rich in terms of biological diversity. India's biological diversity is one of the most significant in the world, since India has only 2% of the total landmass of the world containing about 6% of the world's known wildlife.

The country has 10 different biogeographic zones and 26 biotic provinces.

Table 3.1

S.No.	Biogeographic zones	Biotic provinces
1.	Trans-Himalaya	Ladakh mountains, Tibetan plateau
2.	Himalaya	Northwest, West, Central and East Himalayas
3.	Desert	Thar, Kutch
4.	Semi-arid	Punjab plains, Gujarat Rajputana
5.	Western Ghats	Malabar plains, Western Ghats
6.	Deccan Peninsula	Central highlands, Chotta-Nagpur, Eastern highlands, Central Plateau, Deccan South
7.	Gangetic plains	Upper and Lower Gangetic plains
8.	North-East	Brahmaputra valley, Northeast hills
9.	The Indian Islands	Andaman and Nicobar
10.	Coasts	West and East coast, Lakshadweep

Source: Wildlife Protected Area Network in India: A Review, Wildlife Institute of India, 2000.

1. Trans-Himalayan Region

This area is very cold and arid (4,500 – 6,000 mts. above msl). The only vegetation is a sparse alpine steppe. Extensive areas consist of bare rock and glaciers.

2. Himalayas

The Himalayas consist of the youngest and loftiest mountain chains in the world. The Himalayas have attained a unique personality owing to their high altitude, steep gradient and rich temperate flora.

The forests are very dense with extensive growth of grass and evergreen tall trees. Oak, chestnut, conifer, ash, pine, deodar are abundant in Himalayas. There is no vegetation above the snowline. Several interesting animals live in the Himalayan ranges. Chief species include wild sheep, mountain goats, ibex, shrew, and tapir. Panda and snow leopard are also found here.

3. The Indian Desert

This region consists of parts of Rajasthan, Kutch, Delhi and parts of Gujarat. The climate is characterised by very hot and dry summer and cold winter. Rainfall is less than 70 cms. The plants are mostly xerophytic. Babul, Kikar, wild palm grows in areas of moderate rainfall. Thar desert possesses most of the major insect species. 43 reptile species and moderate bird endemism are found here. The black buck was once the dominant mammal of the desert region, now confined only to certain pockets. The gazelle is the only species of the Indian antelope of which the females have horns. Nilgai the largest antelope of India and the wild ass, a distinct subspecies, is now confined to the Rann of Kutch which is also the only breeding site in the Indian subcontinent for the flamingoes. Other species like desert fox, great Indian bustard, chinkara and desert cat are also found. Camels, wild asses, foxes, and snakes are found in hot and arid deserts.

4. Semi-Arid Areas

Adjoining the desert are the semi-arid areas, a transitional zone between the desert and the denser forests of the Western Ghats. The region includes the arid desert areas of Thar and Rajasthan extending to the Gulf of Kutch and Cambay and the whole Kathiawar peninsula. The natural vegetation is thorn forest. This region is characterized by discontinuous vegetation cover with open areas of bare soil and soil-water deficit throughout the year.

Thorny scrubs, grasses and some bamboos are present in some regions. A few species of xerophytic herbs and some ephemeral herbs are found in this semi-arid tract. Birds, jackals, leopards, eagles, snakes, fox, buffaloes are found in this region. The lion of Gir is the endemic species in this zone.

5. Western Ghats

The mountains along the west coast of peninsular India are the Western Ghats, which constitute one of the unique biological regions of the world.

The Western Ghats extend from the southern tip of the peninsula (8°N) northwards about 1600 km to the mouth of the river Tapti (21 °N). The region covers only 5% of India's land surface but are home to more than about 4,000 of the country's plant species out of which 1800 are endemic. The monsoon forests occur both on the western margins of the ghats and on the eastern side where there is less rainfall. This zone displays diversity of forests from evergreen to dry deciduous.

The mountains rise to average altitudes between 900 and 1500 in above sea level, intercepting monsoon winds from the southwest and creating a rain shadow in the region to their East.

The varied climate and diverse topography create a wide array of habitats that support unique sets of plant and animal species. Apart from biological diversity, the region boasts of high levels of cultural diversity, as many indigenous people inhabit its forests. The Nilgiri langur, lion tailed macaque, Nilgiri tahr, Malabar grey hornbill and most amphibian species are endemic to the Western Ghats.

6. Deccan Peninsula

The Deccan Peninsula is a large area of raised land covering about 43% of India's total land surface. It is bound by the Satpura range on the north, Western Ghats on the west and Eastern Ghats on the east. The elevation of the plateau varies from 900 mts. in the west to 300 mts. in the east. There are four major rivers that support the wetlands of this region which have fertile black and red soil. Large parts are covered by tropical forests. Tropical dry deciduous forests occur in the northern, central and southern part of the plateau. The eastern part of the plateau in Andhra Pradesh, Madhya Pradesh and Orissa has moist deciduous forests.

Fauna like tiger, sloth bear, wild boar, gaur, sambar and chital are found throughout the zone along with small relict populations of wild buffaloes, elephants and barasingha.

7. Gangetic Plain

In the North is the Gangetic plain extending up to the Himalayan foothills. This is the largest unit of the Great Plain of India. Ganga is the main river after whose name this plain is named. The Great Plains cover about 72.4 mha area with the Ganga and the Brahmaputra forming the main drainage axes in the major portion.

The thickness in the alluvial sediments varies considerably with its maximum in the Ganga plains. The physiogeographic scenery varies greatly from arid and semi-arid landscapes of the Rajasthan Plains to the humid and per-humid landscapes of the Delta and Assam valley in the east.

Topographic uniformity, except in the arid Western Rajasthan is a common feature throughout these plains. The plain supports some of the highest population densities depending upon purely agro-based economy in some of these areas. The trees belonging to these forests are teak, sal, shisham, mahua, khair etc. The fauna includes elephants, black buck,

gazelle, rhinoceros, Bengal florican, crocodile, freshwater turtle and a dense waterfowl community.

8. North-East India

North-east India is one of the richest flora regions in the country. The tropical vegetation of northeast India is rich in evergreen and semievergreen rain forests, moist deciduous monsoon forests, swamps and grasslands. It has several species of orchids, bamboos, ferns and other plants. Wild species of cultivated plants such as banana, mango, citrus and pepper are also be found.

Mammalian fauna includes 390 species of which 63% are found in Assam. The area is rich in smaller carnivores. The country's highest population of elephants are found here.

9. The Indian Islands

It is a group of 325 islands: Andaman to the north and Nicobar to the south. The two are separated by about 160 kms by the Ten Degree Channel of the sea. The rainfall is heavy, with both Northeast and Southwest monsoons. About 21 of the 325 islands in the Andaman & Nicobar Islands are inhabited. Many unique plants and animals are found here. About 2,200 species of higher plants are found here of which 200 are endemic. The Andaman & Nicobar Islands have tropical evergreen forests and tropical semi evergreen forests as well as moist deciduous forests, littoral and mangrove forests.

112 endemic species of avifauna, the Andaman water monitor, giant robber crab, 4 species of turtles, wild boar, Andaman day gecko and the harmless Andaman water snake are found only in these islands. The Narcondam hornbill found only in Narcondam is a large forest bird with a big beak. Coral reefs are stretched over an area of 11,000 sq.km. in the Andamans and 2,700 sq.km. in Nicobar.

10. Coasts

India has a coastline extending over 5,500 kms. The Indian coasts vary in their characteristics and structures. The west coast is narrow except around the Gulf of Canbary and the Gulf of Kutch. In the extreme south, however, it is somewhat wider along the south Sahyadri.

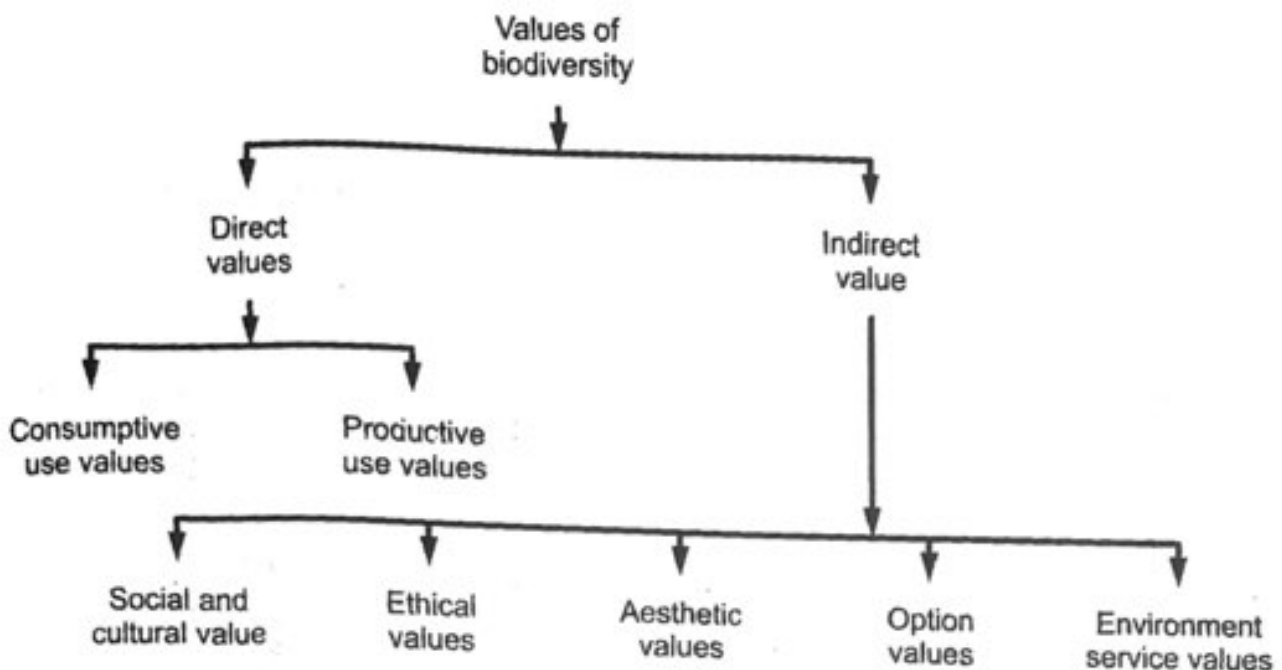
The backwaters are the characteristic features of this coast. The east coast plains, in contrast are broader due to depositional activities of the east-flowing rivers owing to the change in their base levels.

Extensive deltas of the Mahanadi, Godavari, Krishna and Kaveri are the characteristic features of this coast. Mangrove vegetation is characteristic of estuarine tracts along the coast for instance, at Ratnagiri in Maharashtra.

Larger parts of the coastal plains are covered by fertile soils on which different crops are grown. Rice is the main crop of these areas. Coconut trees grow all along the coast.

3.3 VALUE OF BIODIVERSITY

The value of biodiversity is classified into direct and indirect values:



Direct values

The direct value include food resources like grains, vegetables, fruits which are obtained from plant resources and meat, fish, egg, milk and milk products from animal resources. These also include other values like medicine, fuel, timber, fiber, wool, wax, resin, rubber, silk and decorative items.

The direct values are of two types (i) Consumptive use value and (ii) Productive use value.

Consumptive use value: These are the direct use values where the biodiversity products can be harvested and consumed directly. Example: Food, fuel and drugs. These goods are consumed locally and do not figure in national and international market.

(a) Food;

- (i) *Plants:* The most fundamental value of biological resources particularly plants is providing food. Basically three crops i.e. wheat, maize and rice constitute more than two third of the food requirement all over the world.
- (ii) *Fish:* Through the development of aquaculture, techniques, fish and fish products have become the largest source of protein in the world.
- (b) *Fuel:* Since ages forests have provided wood which is used as a fuel. Moreover fossil fuels like coal, petroleum, natural gas are also product of biodiversity which are directly consumed by humans.
- (c) *Drugs and medicines:* The traditional medical practice like ayurveda utilizes plants or their extracts directly. In allopathy, the pharmaceutical industry is much more dependent on natural products. Many drugs are derived from plants like

- (i) **Quinine:** The famous anti malaria drug is obtained from cinchona tree.
- (ii) **Penicillin:** A famous antibiotic is derived from penicillium, a fungus,
- (iii) **Tetracycline:** It is obtained from bacterium.
- (iv) Vinblastin and vincristine, two anti cancer drugs have been obtained from catharanthus plant which has anti cancer alkaloids.

Productive use Values: These are the direct use values where the product is commercially sold in national and international market. Many industries are dependent upon these values. *Example:* Textile, leather, silk, paper and pulp industry etc. Although there is an international ban on trade of products from endangered species like tusks of elephants, wool from sheep, fur of many animals etc. These are traded in market and fetch a booming business.

Indirect Values

Biodiversity provides indirect benefits to human beings which support the existence of biological life and other benefits which are difficult to quantify. These include social and cultural values, ethical values, aesthetic values, option values and environment service values.

Social and cultural value: Many plants and animals are considered holy and sacred in India and are worshipped like Tulsi, peepal, cow, snake etc. In Indian society great cultural value is given to forest and as such tiger, peacock and lotus are named as the national animal, bird and flower respectively.

Ethical: These values are related to conservation of biodiversity where ethical issue of 'all life forms must be preserved' is laid down. There is an existence value which is attached to each species because biodiversity is valuable for the survival of human race. Moreover all species have a moral right to exist independent of our need for them.

Aesthetic value: There is a great aesthetic value which is attached to biodiversity. Natural landscapes at undisturbed places are a delight to watch and also provide opportunities for recreational activities like bird watching, photography etc. It promotes eco-tourism which further generates revenue by designing of zoological, botanical gardens, national parks, wild life conservation etc.

Option values: These values include the unexplored or unknown potentials of biodiversity.

Environment service values: The most important benefit of biodiversity is maintenance of environment services which includes

- (i) Carbon dioxide fixation through photosynthesis.
- (ii) Maintaining of essential nutrients by carbon (C), oxygen (O), Nitrogen (N), Sulphur (S), Phosphorus (P) cycles.
- (iii) Maintaining water cycle and recharging of ground water,
- (iv) Soil formation and protection from erosion,

- (v) Regulating climate by recycling moisture into the atmosphere,
- (vi) Detoxification and decomposition of waste.

3.4 BIODIVERSITY AT GLOBAL, NATIONAL AND LOCAL LEVELS

Global Biodiversity

✓ Humans share the planet earth with as many as 8.7 million different forms of life. Researchers who have analysed the hierarchical categorisation of life on Earth to estimate how many undiscovered species exist say that the diversity of life is not equally divided between land and ocean. Three-quarters of the 8.7 m species - the majority of which are insects - are on land; only one-quarter, 2.2 m, are in the deep, even though 70% of the Earth's surface is water.

Many species are being lost because of human activity, but the magnitude of species lost can only be estimated once we know what species are there. It is reported that, 86% of all plants and animals on land and 91% of those in the seas have yet to be named and catalogued. The conservative estimates by biologists show that the various forms of life on the planet included 7.8 million species of animal, 298,000 species of plant and 611,000 species of mushrooms, mould and other fungi along with 36,400 species of protozoa, single-celled organisms, and 27,500 species of algae.

3.4.1 National Biodiversity

✓ India has a rich biological diversity of flora and fauna. Overall 6% of global species are found in India. It is estimated that India ranks 10th among the plant rich countries of the world, 11th in terms of the number of endemic species of higher vertebrates and 6th among the centre of diversity and origin of agricultural crop.

India has a rich and varied heritage of biodiversity, encompassing a wide spectrum of habitats from tropical rainforests to alpine vegetation and from temperate forests to coastal wetlands.

Of the estimated 5-50 million species of the world's biota; only 1.7 million have been described to date (Groombridge, and Jenkins, 2000), and the distribution is highly uneven. About 7% of the world's total land area is home to half of the world's species, with the tropics alone accounting for 5 million. India contributes significantly to this latitudinal biodiversity trend. With a mere 2.4% of the world's area, India accounts for 7.31% of the global faunal total with a faunal species count of 89,451 species (MoEF, 1999).

India has two major realms called the Palaearctic and the Indo-Malayan, and three biomass, namely the tropical humid forests, the tropical dry/deciduous forests, and the warm desert/semi-deserts.

Some salient features of India's biodiversity are:

- India is one of the 12 centres of origin of cultivated plants.
- India has 5 world heritage sites, 12 biosphere reserves, and 6 Ramsar wetlands.
- Amongst the protected areas, India has 88 national parks and 490 sanctuaries covering an area of 1.53 lakh sq. km.

625
625
1.53
88 national parks
490 Sanctuaries

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625 1.53 88 national parks 490 sanctuaries

3.5 INDIA AS A MEGA-DIVERSITY NATION

India is one of the 17 mega-diversity nations in the world. Mega-diversity, is a term used by an international organization (World Bank Technical paper no. 343). The mega-diverse countries are a group of countries that harbor the majority of the Earth's species and are therefore considered extremely biodiverse. All these countries are located in, or partially in, tropical or subtropical regions. India is located in south Asia, between latitude 6° and 38° N and longitudes 69° and 97° E. Biogeographically, India is situated at the tri-junction of three realms Afro-tropical, Indo-Malayan and Paleo-Arctic realms, and therefore, has characteristic elements from each of them. This assemblage of three distinct realms makes the country rich and unique in biological diversity. The Indian landmass extending over a total geographical area of about 3029 million hectares is bounded by Himalayas in the north, the Bay of Bengal in east, the Arabian Sea in the west, and Indian Ocean in the south. Two areas in India have been identified as megadiversity hot spot areas, which are western ghat forests and eastern Himalayan forests, but India as a whole country is a megadiversity nation. Miller Meier says, 'India is remarkable in both species richness and endemism although it ranks 10th position.

The large species richness and abundance are due to immense variety of climatic and altitudinal condition in country. These vary from the humid tropical Western Ghats to the hot desert of Rajasthan, from cold desert of Ladakh and the icy mountain of Himalayas to the warm coast of peninsular India. The ecosystem diversity is highest in the world, which represents forest ecosystems, grassland ecosystems, wetland ecosystems, coastal and marine ecosystems and the desert ecosystems. There are 10 biogeography zones and 26 biogeography provinces, which are representatives of all the major ecosystems of the world. The country constitutes only 2.4% of the world's land area, but having 11% of flora and 6.5% of fauna of the world. Approximately 65% of the total geographical area has been surveyed so far. Based on this, over 46,000 species of plants which accounts for 15% of the known world plants in this 15000 species of the flowering plants, out of these 35% are endemic and located in 25 endemic centres and 588 genera of monocotyledons 22 are strictly endemic and 81,000 species of animals have been described by the Botanical Survey of India (BSI) established in 1890 and Zoological Survey of India (ZSI) established in 1916, respectively. This list is being constantly upgraded, specially in lower plants and invertebrate animals. India also possesses 40 species of insectivorous plants, 130 species of primitive plants, 130 species of parasites and 70 species of saprophytes. There are about 15000 species of angiosperms, 65 species of gymnosperms, 1232 species of pteridophytes, 2850 species of bryophytes, 6990 species of algae, 2075 species of lichens, 14500 species of fungi and more than 850 species of virus and bacteria are found in India. It contains 1,27,000 species of the world of which more than 45,000 species are plants and more than 80,000 species are animals. India contains about 75000 animal species out of which 80% are insects. In

animals 62% of amphibians and 32% of reptiles are endemic. There is no clear estimation about marine biota in Indian sub-continent. Long coast line have abundance of fish, crustaceans, corals, reptiles, seaweeds, molluscs and mammals. In India, mangroves occur along the coast-line comprising a total area of 6740 sq.kms; which is nearly 12% of the world's mangrove areas. The country also has four Coral Reef areas located in Gulf of Mannar, Gulf of Kutch, Lakshadweep and Andaman and Nicobar Islands. The Western Ghats in peninsular India, which extend in the southern states are a treasure house of species diversity and has about 5000 species. It is estimated that almost one third of animals varieties found in India are in Western Ghat of Kerala alone. The country is also one of the 12 primary centres of origin of cultivated plants and domesticated animals. It is considered to be the homeland of 167 important plant species of cereals, millets, fruits, condiments, vegetables, pulses, fibre crops and oilseeds, and 114 breeds of domesticated animals.

India contains a great wealth of biological diversity in its terrestrial and aquatic areas. This richness is shown in absolute numbers of species and the proportion they represent of the world total.

Table 3.2 Comparison between the Number of Species in India and the World

Group	Number of species in India (SI)	Number of species in the world (SW)	SI/SW (%)
Mammals	350	4629	7.6
Birds	1224	9702	12.6
Reptiles	408	6550	6.2
Amphibians	197	4522	4.4
Fishes	2546	21730	11.7
Flowering Plants	15000	250000	6.0

3.6 HOT-SPOTS OF BIODIVERSITY

Biodiversity hotspots are areas that support natural ecosystems that are largely intact and where native species and communities associated with these ecosystems are well represented. These are also areas with a high diversity of locally endemic species, which are species that are not found or are rarely found outside the hotspot.

A biodiversity hotspot is a biogeographic region with a significant reservoir of biodiversity that is under threat from humans. The concept of biodiversity hotspots was originated by Norman Myers, a British environmentalist, specialising in biodiversity. To qualify as a biodiversity hotspot on the hotspot-map, a region must meet two strict criteria: it must contain at least 0.5% or 1,500 species of vascular plants as endemics, and it has to have lost at least 70% of its primary vegetation. Around the world, 25 areas qualify under this definition, with nine other possible candidates.

Source: Myers, N. et. al., 2000. Biodiversity hotspots for conservation priorities. Nature 403:853-858

These sites support nearly 60% of the world's plant, bird, mammal, reptile, and amphibian species, with a very high share of endemic species.

India is one of the richest countries in the world in terms of biodiversity. It lies within the Indo-malaya eco-zone and completely houses two of the thirty-four biodiversity hotspot in the world. The two hotspots are: The Eastern Himalayas and the Western Ghats. The third, Indo-Burma, lies partially within the Indian North-East. There is a huge species diversity in India, with several of the species being endemic to their native ranges in India.

Eastern Himalayas

Phytogeographically, the Eastern Himalayas form a distinct floral region. The area comprises Nepal, Bhutan, and neighbouring states of northern India, along with a continuous sector of the Yunnan province in southwest China. Although all Himalayan forests lie well north of the Tropic of Cancer, and some of them are at altitudes of 1780 to 3500 metres, they can be considered tropical forests since they occur largely within the climatic tropics.

It is believed that forest cover in the Eastern Himalayas has dwindled from 340,000 km² to 110,000 km² with a mere 53,000 km² of primary forests. Despite this loss, the north-eastern region is home to some botanical rarities. One of these is the *Sapria Himalayana*, a parasitic angiosperm that has been sighted only twice since 1836.

This region is the meeting ground of the Indo-Malayan and Indo-Chinese biogeographical realms as well as the Himalayan and Peninsular Indian elements, formed when the peninsular plate struck against the Asian landmass, after it broke off from Gondwana land.

- The Eastern Himalayas display an ultra-varied topography, a factor that fosters species diversity and endemism.
- Many deep and semi-isolated valleys are exceptionally rich in endemic plant species. In Sikkim, in an area of 7298 km², of the 4250 plant species, 2550 (60%) are endemic (Myers, 1988). In India's sector of the area, there are about 5800 plant species, of which roughly 2000 (36%) are endemic.
- In Nepal, there are around 7000 plant species, many of which overlap with those of India, Bhutan, and even Yunnan. Of these species, at least 500 (almost 8%) are believed to be endemic to Nepal.
- Bhutan possesses an estimated 5000 species, of which as many as 750 (15%) are considered to be endemic to the Eastern Himalayas.

Western Ghats

Of India's 49,219 plant species, 1600 endemics (40% of the total number of endemics) are found in a 17,000 km² strip of forest along the seaward side of the Western Ghats in Maharashtra, Karnataka, Tamil Nadu, and Kerala (WCMC, 1992). Forest tracts up to 500 metres in elevation, comprising one-fifth of the entire forest expanse, are mostly evergreen, while those in the 500-1500 metres range are semi-evergreen. There are two main

centres of diversity, the Agastyamalai Hills and the Silent Valley/New Amambalam Reserve basin (Myers, 1988).

Very little has been documented recently about the status of the forest cover, except that it seems to have declined between 1972 and 1985 at a rate paralleling that for India as a whole, which implies a loss of over 2.4% annually. If we extrapolate from 1986 to 1989, this means a total loss of almost 34% for 1972 to 1989. According to Myers et al, (2000), only 6.8% of the original extent of vegetation exists today. Still worse is the decline of the primary forest: the amount remaining seems to be no more than 8000 km². All but isolated pockets of original forest have been opened up by shifting cultivation, allowing a take-over by deciduous species and bamboo among other forms of 'degenerate' vegetation.

3.7 THREATS OF BIODIVERSITY

Biodiversity is under serious threat as a result of human activities. The main dangers worldwide are population growth and resource consumption, climate change and global warming, habitat conversion and urbanisation, over-exploitation of natural resources and environmental degradation.

Table 3.3 The table below summarise the main threats, a country, usually faces:

Main threats	Some underlying causes
<i>Threats in terrestrial areas:</i>	
Degradation, destruction and natural habitats	Spread of the urbanised areas, road network and industrial areas and associated problems (noise, pollution); abandon of former agricultural practices that were favourable to biodiversity
Decrease in the capacity of the agricultural areas to host wildlife	Intensification of agricultural practices (yielding pollution and disturbance) and disappearance of landscape elements that provide food and shelter that are exploitable by wildlife (such as hedges, trees, ponds, etc.)
Pollution of soils, air and water	Excess of heavy metals (industry, roads), manure and pesticides (agriculture) and other pollutants
Invasions by alien species	International trade and transport (roads, railways, rivers), gardening practices, exotic trees in forestry, exotic pests released in the wild, climate change, etc.
Epidemics affecting wildlife	Arrivals of pathogens that are favoured by the introduction of exotic species, pollution and the destruction of habitats
Climate change	Carbon emissions, deforestation and other land use changes due to human activities
Dessication of soils and wetlands	Excess pumping of underground water tables

Main threats	Some underlying causes
Recreation and leisure	Overuse of green open spaces and wild areas, little respect for nature, mountain biking and motor sports in fragile areas, dogs not on leash
<i>Threats in marine areas:</i>	
Overfishing and decline of species	Industrial fishing, over exploitation of target species, by-catch
Pollution and eutrophication	Land-based activities (river run-off), atmospheric deposition, maritime traffic
Degradation and destruction of the sea floor	Beam trawling, dredging, sand and gravel extraction
Alien species introduction	Maritime trade (ballast waters, fouling), leisure navigation, mariculture, climate change
Leisure and tourism	Coastal development, water quality in summer (high population), mechanical beach cleaning, noise and other perturbations due to the high population

India's biodiversity is one of the most significant in the world. However, this rich biodiversity of India is under severe threat owing to habitat destruction and over exploitation. The country has several problems such as overpopulation, large number of cattle heads, growing demand for land, energy and water supply. Unplanned developmental works and over exploitation of resources have made its living resources most vulnerable. The major causes and issues related to threat to biodiversity are discussed below:

Habitat Loss

Habitat loss is one of the biggest threats to biodiversity and the largest reason for species going extinct. Cutting forests to create fields, filling in wetlands to build houses, and creating dams that change river flow are all examples of habitat destruction. The rapidly growing human population is putting more and more pressure on existing habitats. In most of the countries, original forest cover is lost and converted to farmlands. This has ecological consequences and many countries now have national parks, sanctuaries and other ways of preserving land for future use. In India, several rare butterfly species are facing extinction with the fast rate of habitat destruction in the Western Ghats.

Poaching of Wild Life

Another major problem is the poaching and killing of animals and birds. Many species of animals and birds are hunted for their flesh, skins, furs, feathers, bones, tusks, horns, and venoms and so on. This unjustified killing is going on even today. Apart from poachers and smugglers, various

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ethnic groups or Adivasis kill faunal wealth for satisfying their faith, taboo or for amusement. In addition, the clearing of natural forests to make room for commercial plantations causes innumerable damage to wildlife.

In the first half of the century, lion and in particular tiger have been decimated by the British and the Indian royalty for purposes of sport. At present tiger is being killed on account of the sale of its bones and other parts to meet the unprecedented demand for traditional Chinese medicine. The illegal poaching and unauthorized hunting has harmed the fauna to such an extent that many species of animals have become extinct and many are on the verge of extinction. At least 30% of the trade is illegal and stands next only in value to narcotics trade. As long as this type of money is involved, rare animals will continue to be killed illegally.

There is a conflict between biodiversity conservators and general public because the public is not taking interest in wildlife management. A general awareness has to be developed. People's participation and more stringent implementation of anti poaching laws and better backup facilities like faster vehicles, modernised weapons and more power to tackle poachers at the field level can perhaps reduce the magnitude of the problem of indiscriminate poaching and extermination of many rare species.

Man-Wildlife Conflict

When wild animals leave the protected areas (forests) to raid human settlements in search of food and water it gives rise to a conflict between man and wildlife. The main reason for this conflict is the growing anthropogenic pressure on wildlife habitat which results in:

- Fragmentation and honeycombing of animal habitat.
- Loss of corridors and migratory routes for long-range animals such as elephants, big cats (tigers, leopards, bears) besides others.
- Loss of food and water in their habitat due to the shrinking of forest cover and loss of biodiversity.

When wild animals destroy crops causing economic and food losses to farmers, affect water supplies, kill or injure humans and cause havoc in the lives of human beings, they retaliate by killing the wild animal.

The conflict is fast becoming a critical threat to the survival of many globally endangered species such as the Sumatran tiger, Asian Lion, Snow Leopard, Red Colobus Monkey and many more. It has also been observed that the more volatile species are more prone to extinction because of injury and death caused by humans, traffic (road, railway track) or other accidents (for example, falling into traps, wells, poisoning, electric fences and so on).

Considering that this conflict will always remain, strategies are being evolved by government wildlife managers, scientists and local communities, not just for the protection of humans but also for the conservation of biodiversity (wildlife).

This multi-faceted problem can be minimized with good management practices and approaches involving low-cost strategies such as electric

fencing, community-based natural resource management schemes, incentives, and insurance programmes along with regulated harvesting and wildlife or human translocation.

CASE STUDY

Man-Wildlife Conflict

In India, people living in and around the protected areas mainly depend on forest products, agriculture and agro-pasture. As a result, very often man-wildlife conflicts result in crop loss, injury or loss of human life and sometimes the death of wild animals.

About 1,07,770 people live in 117 villages in and around the Sariska Tiger Reserve Project in Rajasthan. Agriculture and rearing livestock are the main sources of livelihood for them. Many species of wild herbivores such as the Nilgai and wild boars are to be blamed for 50 per cent of the damage to their crops while sambar, Chital, the common langur, rhesus monkeys and parakeets are blamed for the rest. Wild carnivores such as tigers and leopards are responsible for livestock loss. Tigers prey on big domestic animals like cattle and buffalos while leopards prey on goats, sheep and calves.

3.8 ENDANGERED AND THREATENED SPECIES OF INDIA

A species is said to be **extinct** when it is not seen in the wild for fifty years at a stretch like dodo, passenger pigeon. A species is said to be in **vulnerable** category if its population is facing continuous decline due to overexploitation or habitat destruction. Such species are still abundant but under a serious threat of becoming **endangered** if causal factors are not checked. Species those are not endangered or vulnerable at present, but are at a risk, are categorised as **rare** species. These species are usually localised within restricted area i.e., they are usually endemic. Sometimes they are thinly scattered over a more extensive area.

A species is said to be endangered when its number has been reduced to a certain level or whose habitats have been drastically reduced and if such species is not protected and conserved, it is in immediate danger of extinction. The International Union for Conservation of Nature and Natural Resources (IUCN) publishes the Red Data Book which includes the list of endangered species of plants and animals. The red data symbolises the warning signal for those species which are endangered and if not protected are likely to become extinct in near future. Critically Endangered species in India According to the Red Data Book of IUCN, there are 47 critically endangered species in India. The Red list at Rio +20 Earth Summit, 2012, has listed 132 species of plants and animals as Critically Endangered from India.

Botanical Survey of India (BSI) has published a Green Book which lists rare plant species growing in protected areas like botanical gardens. In India, nearly 450 plant species, 150 mammals and 150 species of birds are estimated to be threatened and identified in the categories of endangered or rare while an unknown number of species of insects are endangered.

Few of the endangered mammals, reptiles, amphibians, fishes, birds and spiders are listed below:

Mammals: The Pygmy hog, Himalayan wolf, Kashmir stag, Javan Rhinoceros, red fox, sloth bear, red panda, tiger, leopard, striped hyena, Indian lion, golden cat, desert cat.

Reptiles: Gharial, green sea turtle, leatherback turtle, Sispara day gecko, hawksbill turtle.

Amphibians: Kerala Indian frog, White-spotted bush frog, Munnar bush frog, Tiger toad.

Fishes: The Ganges Shark, large-tooth sawfish, knife-tooth sawfish

Birds: Great Indian Bustard, Siberian Crane, Bengal Florican, Forest owl, Pink-headed duck, Himalayan Quail, Indian Vulture, Red-headed vulture.

Spiders: Rameshwaram Ornamental Spider, Gooty Tarantula.

3.9 ENDEMIC SPECIES OF INDIA

An endemic species is one whose habitat is restricted to a particular area. The term could refer to an animal, a plant, a fungus, or even a microorganism. The definition differs from "indigenous," or "native," species in that the latter, although it occurs naturally in an area, is also found in other areas. Endemic species are often endangered, and particular examples may become a focus point for campaigns to protect biodiversity in a given environment. Some have become national, or regional, emblems.

There are two ways in which a species may become endemic to a particular area. An initially widely distributed population may disappear from many of its habitats, due to changes which have occurred. These could be climate changes, an influx of predators, or human activities. Eventually, the organism may be confined to just one area: this type is known as a *paleo-endemic species*. Alternatively, various factors could cause two populations of a given species to become isolated from one another. For example, as a result of plate tectonics, a continent may split apart, forming two new continents, each with its own population of a given organism. Over long periods of time, these two populations evolve differently, because they cannot interbreed with one another, and eventually they are sufficiently different from one another to be classified as separate species. These are known as *neo-endemic species*.

India has many endemic plant and animal species. Among plants, species endemism is estimated at 33%. More than 140 endemic genera exist, but there are no endemic families (Botanical Survey of India, 1983). Areas rich in endemic species are north-east India, the Western Ghats and the

north-western and eastern Himalayas. A small pocket of local endemism also occurs in the Eastern Ghats. The Gangetic plains are generally poor in endemics, while the Andaman and Nicobar Islands contribute at least 220 species to the endemic flora of India (Botanical Survey of India, 1983). In India, the locations recognised for conservation action are the Agastyamalai Hills, Silent Valley and New Amarambalam Reserve and Periyar National Park (all in the Western Ghats), and the Eastern and Western Himalaya. Endemism among mammals and birds is relatively low. Only 44 species of Indian mammals have a range that is confined entirely within Indian territorial limits. Four endemic species of conservation significance occur in the Western Ghats. They are the Lion-tailed macaque *Macaca silenus*, Nilgiri leaf monkey *Trachypithecus johni*, Brown palm civet *Paradoxurus jerdoni* and Nilgiri tahr *Hemitragus hylocrius*. Only 55 bird species are endemic to India, with distributions concentrated in areas of high rainfall. They are located mainly in eastern India along the mountain chains where the monsoon shadow occurs, south-west India and the Nicobar and Andaman Islands. In contrast, endemism in the Indian reptilian and amphibian fauna is high. There are around 214 endemic reptiles, and 110 endemic amphibian species.

3.10 BIODIVERSITY CONSERVATION

Biodiversity is a wealth to which no value can be put. The survival of the human race is dependent on conservation of biodiversity. It is evident that this invaluable heritage is being destroyed at an alarming rate due to several reasons. Measures are being taken up at national and international levels to address this issue. The Earth Summit produced a plan of action on a number of issues (Agenda 21) including conservation of biodiversity during the 21st century.

There are several strategies which are adapted for conservation of Biodiversity. Some of these are:

1. Legislation

Formal policies and programmes for conservation and sustainable utilisation of biodiversity resources dates back to several decades. The concept of environmental protection is enshrined in the Indian constitution in articles 48a and 51a(g). Major central acts relevant to biodiversity include:

- Environment Protection Act, 1986
- Fisheries Act, 1897
- Forest Act, 1927
- Forest (Conservation) Act, 1980
- Wildlife (Protection) Act 1972 and Wildlife (Protection) Amendment Act 1991

2. In-situ Conservation

Conserving the animals and plants in their natural habitats is known as *in situ* conservation. The established natural habitats are:

- National parks and sanctuaries
- Biosphere reserves
- Nature reserves
- Reserved and protected forests
- Preservation plots
- Reserved forests

The first such initiative was the establishment of the Corbett National Park in 1936. National Parks are highly protected by law. No human habitation, private land holding or traditional human activity such as firewood collection or grazing is allowed within the park. Sanctuaries are also protected but certain types of activities are permitted within these areas.

Biosphere Reserves are another category of protected areas. Under this, a large area is declared as a Biosphere Reserve where wildlife is protected, but local communities are allowed to continue to live and pursue traditional activities within the Reserve. The Government of India has set up seven biosphere reserves: Nokrek (Meghalaya), Nilgiri (Karnataka, Kerala, Tamil Nadu), Namdapha (Arunachal Pradesh), Nanda Devi (Uttar Pradesh), Sundarbans (West Bengal), Great Nicobar (Andaman & Nicobar Islands), Gulf of Mannar (Tamil Nadu).

Several special projects have also been launched to save certain animal species which have been identified as needing concerted protection effort. These projects are designed to protect the species in situ, by protecting and conserving their natural habitat. Project Tiger, Project Elephant, Save the Barasingha campaign are examples of this initiative. Other strategies include offloading pressure from reserve forests by alternative measures of fuelwood and fodder need satisfaction by afforestation of degraded areas and wastelands.

A programme "Eco-development" for *in-situ* conservation of biological diversity involving local communities integrates the ecological and economic parameters for sustained conservation of ecosystems by involving local communities with maintenance of earmarked regions surrounding protected areas.

Approximately, 4.2% of the total geographical area of the country has been earmarked for extensive *in-situ* conservation of habitats and ecosystems. A protected area network of 85 national parks and 448 wildlife sanctuaries has been created. This network have been significant in restoring viable population of large mammals such as tiger, lion, rhinoceros, crocodiles and elephants.

3. Ex-situ Conservation

Ex-situ conservation of plants and animals preserve/or protect them away from their natural habitat. This could be in zoological parks and botanical gardens or through the forestry institutions and agricultural research

centres. A lot of effort is under way to collect and preserve the genetic material of crops, animal, bird and fish species. This work is being done by institutions such as the National Bureau of Plant Genetic Resources, New Delhi, the National Bureau of Animal Genetic Resources, etc. Reproduction of an animal or plant into the habitat from where it has become extinct is another form of *ex situ* conservation. For example, the Gangetic *gharial* has been reintroduced in the rivers of Uttar Pradesh, Madhya Pradesh and Rajasthan where it had become extinct. Seed banks, botanical, horticultural and recreational gardens are important centres for *ex-situ* conservation. *Ex-situ* conservation measures complement in-situ conservation.

4. Recording Indigenous Knowledge

The lives of local communities are closely interwoven with their environment, and are dependent upon their immediate resources for meeting their needs. These communities have a vast knowledge about local flora and fauna which is very important for biodiversity conservation. Much of this knowledge is orally passed on from generation to generation. Such indigenous knowledge needs to be recorded and preserved before it is lost. Several organizations have recognized this and are working to record the knowledge and preserve it for posterity.

5. Community Participation in Biodiversity Conservation

It is being recognized that no legal provisions can be effective unless local communities are involved in planning, management and monitoring conservation programmes. There are several initiatives to do this, both by government as well as non-governmental organizations. For example, the Joint Forest Management philosophy stresses involvement of village communities in regenerating and protecting degraded forest land in the vicinity of villages. Successful conservation strategies will have to have the confidence and participation of the local communities.

6. International Conservation Strategies

Conserving biodiversity is not an issue confined to any one country or community. It is a crucial global concern. Several international treaties and agreements are in place in the attempt to strengthen international participation and commitment towards conserving biodiversity. Some of these are:

- **The Convention on Biological Diversity:** This was signed during the Earth Summit in 1992. It focuses not only on conserving biodiversity but also on sustainable use of biological resources and equitable sharing of benefits arising from its use.
- **The Convention on International Trade in Endangered Species of Wild Flora and Fauna (CITES):** This is an international treaty which is designed to protect wild plants and animals affected by international trade. The treaty, in force since 1975, controls the export, import and re-export of endangered and threatened wildlife.

- **The Convention on Wetlands of International Importance:** This Convention, also known as the Ramsar Convention, was signed in Ramsar (Iran) in 1971 and came into force in December 1975. It provides a framework for international cooperation for the conservation of wetland habitats which have been designated to the 'List of Wetlands of International Importance'.

Saving biodiversity - Saving life

Following are some conservation actions that we can take up:

- Plant trees. Grow native species of plants (trees, shrubs and climber) where possible, this would attract local wildlife such as birds, butterflies and insects.
- Grow local vegetables in your school garden that are not usually available in the markets. This would allow help conserve them for generations to come.
- Initiate, organize and participate in responsible citizen action against existing or proposed activities that harm or are likely to harm local biodiversity.
- Make a list of different kinds of trees in your campus or locality. For each one find out the names, uses, flowering season, animals and birds that depends on it. Present this information in an interesting way, and put it up by the tree. Many people will stop by to read this information and know more about the tree.
- Curb our greed for products made out of animal parts like skin, fur, ivory, bones, nails, etc., to discourage wildlife traders and poachers, and spare the lives of the remaining animals.
- Adopt vegetarianism which would require fewer animals to be fattened for slaughtering and more plants to be grown for food.
- Avoid using insecticides, pesticides and inorganic fertilizers and try to use natural plant-based substitutes wherever possible. Paper and cloth should replace non-biodegradable plastic and polyester which damage the ecosystem.
- Make children aware of their surroundings and the need for biodiversity.
- Promote bio-farming which is less intensive and environmental-friendly.
- Make use of sustainable technologies like smokeless chulhas, ground water recharging unit, wind energy, solar power, etc.
- Set up 'Community Sanctuaries' for free ranging animals, migratory birds and endemic species. Establish voluntary 'Village Reserves' for plants and animals.
- Assist National level Bodies in recording and preserving rare and endemic species
- Create 'Biodiversity Registers' in communities, schools, villages.

EXERCISES

1. What is meant by biodiversity? Describe the uses and importance of biodiversity.
2. Explain the levels of biodiversity.
3. Discuss various biogeographic zones of India.
4. What are the hotspots of biodiversity? Discuss the hotspots of India.
5. Write a short note on threats to biodiversity.
6. Discuss the consumptive and productive values of biodiversity.
7. What are extinct, endangered, vulnerable and rare species? Discuss giving examples.
8. How are poaching and human-wildlife conflicts responsible for depletion of biodiversity?
9. What do you understand by endemic species? Give examples of the endemic species of India.
10. What are the various approaches for conservation of biodiversity? Discuss each in detail.
11. Comment upon Indian biodiversity with special reference as a mega diverse nation.
12. Write short notes on: (i) Red data book, (ii) Habitat loss (iii) Endangered species

